

A Statistical Theory of Daily-Bar Volatility under a Noisy Open

The theory section and Appendix B of
When the Open Overreacts: Measuring Daily Volatility in Frontier Markets without Options

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Working paper, October 2026

Abstract

The daily bar's opening price can carry a transient error that the session undoes, as it does in Nepal's auction-set open. This document develops the statistics of that problem and states eight propositions, with complete proofs in Appendix B. It shows what the unbiasedness coefficient $b = \mathbb{E}[or]/\mathbb{E}[o^2]$ identifies, with a sharp lower bound on the error that replaces a weaker published one. It derives exactly how much of the error each classical range estimator absorbs, and how a price band inflates b , with a closed-form factor. It explains why one coefficient pooled across securities whose opens differ penalises the estimator that reads the open. Finally, it shows how a rolling calibration lags a rule change, and when lagged realised measures identify an estimator's calibration. A seventh proposition, added after an independent audit, shows what Anam's estimator does with a noisy open. Its overnight term understates the efficient overnight variance unless the error is proportional to the news. Its extended range keeps any error that set the high or the low. Its open-free form still absorbs part of the error's variance: between 0.25 and 0.38 of it at the error sizes the data imply, against about half for Parkinson. Each closed form, identity and inequality was checked against simulation or numerical integration by `scripts/45_theory_checks.py`. That ledger has 187 checks, and 187 pass. Every step of every proof was then verified by `scripts/46_theory_proofs.py`, symbolically with SymPy wherever the step is algebra or calculus. Of its 321 steps, 321 pass. The seventh proposition's closed forms were checked by `scripts/48_kernel_theory.py` (63 checks, 63 passing). An eighth proposition explains why a single coefficient fails. In a panel, the open's reliability is a weighted mean of two parts: a market part that the session keeps almost entirely, and a stock-specific part that it mostly reverses. The market's move can be read from the other securities' opens, free of the security's own error. Its checks are in `scripts/54_market_open_theory.py` (35 checks, 35 passing). The applications to the NEPSE data and the one new test were fixed in advance in plan M19.

1 A statistical theory of the daily bar under a noisy open

Sections 6.6–6.8 of the paper report four facts:

- NEPSE's open is mostly reversed during the session;
- a wider pre-open band lowered the unbiasedness coefficient sharply;
- an estimator that ignores the open forecast better than one that reads it, in every frontier panel;
- a pooled calibration lagged the band reform.

This section gives the statistics behind those facts in six propositions. A seventh, added after an independent audit, says what Anam's estimator does with a noisy open. An eighth, derived during the development of Anam II (plan M22), says what the cross-section of opens can recover.

Proofs are in Appendix B. The script `scripts/45_theory_checks.py` compared each closed form, identity and inequality with a simulation or a numerical integral, as plan M19 required. The plan was registered before the script was written. Table 6 lists the checks. A second script verified every step of every proof, symbolically wherever the step is algebra or calculus (Table 7). The propositions are post hoc relative to plans M14–M18, because they were derived after those results were known. Only the test of Section 1.5 had its prediction fixed in advance. Proposition 7 was checked by a third script, `scripts/48_kernel_theory.py`.

1.1 The model

For session t , let C_- be the printed previous close and O, H, L, C the printed open, high, low and close. The bar's coordinates are:

- the overnight return $o = \ln(O/C_-)$;
- the intraday return $c = \ln(C/O)$;
- the close-to-close return $r = o + c$;
- the high and low relative to the open, $u = \ln(H/O)$ and $d = \ln(L/O)$;
- the range $R = u - d$.

Following Appendix A of the paper, write

$$o = e_o + \eta - \varepsilon_-, \quad c = e_c - \eta + \varepsilon, \quad r = e_o + e_c + \varepsilon - \varepsilon_-. \quad (1)$$

Here e_o and e_c are the efficient overnight and intraday returns, η is the open's transient error, and ε and ε_- are the errors of today's and the previous close. Two combinations do all the work:

- $o^* = e_o - \varepsilon_-$ is the efficient open measured from the printed previous close;
- $x = e_c + \varepsilon$ is everything that happens after the open.

Then $o = o^* + \eta$ and $r = o^* + x$. All moments are raw, as in

$$b = \frac{\mathbb{E}[or]}{\mathbb{E}[o^2]}. \quad (2)$$

This is the unbiasedness coefficient of the price-discovery literature (Biais et al., 1999; Barclay and Hendershott, 2003).

Assumption 1 (Nothing known at the open predicts the rest of the session). Second moments are finite, $\mathbb{E}[o^2] > 0$, and $\mathbb{E}[o^*x] = \mathbb{E}[\eta x] = 0$.

Assumption 1 holds if efficient prices are martingales with respect to information that includes the open's error, and if the closing error is not predictable from the open. It places no restriction on how the error relates to overnight news. The open may overreact ($\mathbb{E}[o^*\eta] > 0$), underreact ($\mathbb{E}[o^*\eta] < 0$) or be stale.

Assumption 2 (A Brownian session with an instantly corrected open).

- The closing errors are zero.
- After the printed open $o = o^* + \eta$, the price follows the efficient path $o^* + \sigma_c W(s)$, $s \in (0, 1]$, where W is a standard Brownian motion independent of (o^*, η) .
- The high and the low are the extremes of the printed path, $\ln(H/C_-) = \max(o, o^* + \sigma_c M)$ and $\ln(L/C_-) = \min(o, o^* + \sigma_c m)$, where $M = \sup W$ and $m = \inf W$.
- The close is efficient, so $x = \sigma_c D$ with $D = W(1)$.

Where the error's law matters, $\eta \sim N(0, \sigma_\eta^2)$ independently of (o^*, W) . The error's scale relative to the session is $\rho = \sigma_\eta / \sigma_c$.

Assumption 3 (A band on the open). The auction would print a latent open o_L , with $\mathbb{E}[r \mid o_L] = b_L o_L$. The printed open is $o = g(o_L) = \max\{-\beta, \min(\beta, o_L)\}$, and the band leaves the close unaffected.

A linear conditional mean holds for every elliptical law of (o_L, r) , Gaussian or Student- t (Cambanis et al., 1981). Assumption 3 therefore allows heavy tails.

1.2 What the unbiasedness coefficient identifies

Proposition 1 (Identification). *Under Assumption 1:*

- (a) $\mathbb{E}[or] = \mathbb{E}[oo^*]$, so $b = \arg \min_\gamma \mathbb{E}[(o^* - \gamma o)^2]$: bo is the best linear predictor of o^* through the origin (all moments here are raw). The effective open $C_{\exp}(bo)$ therefore predicts the efficient open on the log scale. If $\mathbb{E}[o^* \mid o]$ is linear in o , then $bo = \mathbb{E}[o^* \mid o]$.
- (b) $\mathbb{E}[oc] = -(1 - b) \mathbb{E}[o^2]$: the session reverses the fraction $1 - b$ of the overnight move. This part uses only the definition of b .
- (c) $\mathbb{E}[\eta^2] = (1 - b)^2 \mathbb{E}[o^2] + \mathbb{E}[(o^* - bo)^2]$. Hence

$$\mathbb{E}[\eta^2] \geq (1 - b)^2 \mathbb{E}[o^2], \quad (3)$$

with equality if and only if $o^* = bo$ almost surely.

(d) Three special cases:

- If $\mathbb{E}[o^* \eta] = 0$, then $b = \mathbb{E}[o^{*2}] / (\mathbb{E}[o^{*2}] + \mathbb{E}[\eta^2])$ and $\mathbb{E}[\eta^2] = (1 - b) \mathbb{E}[o^2]$.
 - If $\eta = \theta o^* + \nu$ with $\mathbb{E}[o^* \nu] = 0$, then $b = (1 + \theta) \mathbb{E}[o^{*2}] / \{(1 + \theta)^2 \mathbb{E}[o^{*2}] + \mathbb{E}[\nu^2]\}$.
 - If the open moves only part of the way, $o = \lambda o^*$, then $b = 1/\lambda$.
- (e) The second moments of (o, r) identify b , $\mathbb{E}[o^2]$ and $\mathbb{E}[r^2] = \mathbb{E}[o^{*2}] + \mathbb{E}[x^2]$. They do not identify how $\mathbb{E}[r^2]$ divides between the open and the session. Every value of $\mathbb{E}[o^{*2}]$ in $[b^2 \mathbb{E}[o^2], \mathbb{E}[r^2]]$ is attained by some law satisfying Assumption 1 with the same Gaussian law of (o, r) . Correspondingly, $\mathbb{E}[\eta^2]$ ranges over $[(1 - b)^2 \mathbb{E}[o^2], (1 - b)^2 \mathbb{E}[o^2] + \mathbb{E}[r^2] - b^2 \mathbb{E}[o^2]]$.

Part (c) is a variance decomposition. The error's second moment is the reversal the data show, $(1 - b)^2 \mathbb{E}[o^2]$, plus the part of the efficient open that the printed open cannot predict. The decomposition yields the sharp form of the bound given in Appendix A of the paper. That bound, $\mathbb{E}[\eta^2] \geq \mathbb{E}[o^2] \{(\sqrt{5 - 4b} - 1)/2\}^2$, is valid, but it is strictly smaller than (3) for every $b < 1$. At $b = 0.3$ the sharp bound is 2.17 times larger. Appendix A also states that its bound is attained when the error is perfectly correlated with news. That is incorrect. Under perfect correlation, $\eta = \{(1 - b)/b\} o^*$, and the error's second moment equals the sharp bound, $(1 - b)^2 \mathbb{E}[o^2]$. The correction is registered as M-026.

Part (e) is why the paper sets the level of every estimator on the close-to-close scale. The only level the bar identifies without assumptions about the error is $\mathbb{E}[r^2]$. How it divides between the overnight move and the session is identified only up to the interval in (e). The overnight term of Anam's estimator, $(bo)^2$, has expectation $b^2 \mathbb{E}[o^2]$, the lower end of that interval. That lower end is exact under pure proportional overreaction, and Proposition 7(a) says what it misses otherwise. Part (d) gives another reading of the S&P 500's pooled b of 2.21 (Section 6.8): it is what an index open reflecting about 45% of the overnight move would produce.

Table 1: The open's error as a share of the open-to-close proxy $\mathbb{E}[c^2]$ (Proposition 1(c)), with 95% joint-bootstrap intervals (plan M15's bootstrap, 499 replicates).

Sample	b	$\mathbb{E}[o^2]/\mathbb{E}[c^2]$	Published bound $\{\frac{\sqrt{5-4b}-1}{2}\}^2 \frac{\mathbb{E}[o^2]}{\mathbb{E}[c^2]}$	Sharp bound $(1-b)^2 \frac{\mathbb{E}[o^2]}{\mathbb{E}[c^2]}$	Independence $(1-b) \frac{\mathbb{E}[o^2]}{\mathbb{E}[c^2]}$
NEPSE, A1	0.307	0.225	0.050 [0.038, 0.065]	0.108 [0.078, 0.145]	0.156 [0.126, 0.197]
NEPSE, B	0.216	0.293	0.078 [0.064, 0.094]	0.180 [0.144, 0.224]	0.230 [0.197, 0.268]
NEPSE, A2	0.356	0.274	0.054 [0.032, 0.074]	0.114 [0.059, 0.166]	0.176 [0.118, 0.224]
NEPSE, C	0.130	0.552	0.172 [0.139, 0.198]	0.417 [0.323, 0.494]	0.480 [0.397, 0.543]
NIFTY 50	0.929	0.505	0.002 [0.000, 0.008]	0.003 [0.000, 0.010]	0.036 [0.003, 0.071]

Application (plan M19, part B1). Table 1 recomputes the bound with the joint bootstrap of plan M15, over the rows of that plan's post hoc table. Under the $\pm 2\%$ band in the first regime, the sharp bound puts the open's error at no less than 10.8% of the open-to-close proxy, with 95% interval [0.078, 0.145]. This is 2.16 times the published bound of 5.0%. Under independence the share is 15.6%. After the band was widened (regime C), the sharp bound is 41.7%. On the NIFTY 50, $b = 0.929$, and the bound is 0.003 with interval [0.000, 0.010], which is indistinguishable from zero.

1.3 What the classical estimators measure

Write the classical estimators as follows:

- Parkinson's, $P = R^2/(4 \ln 2)$ (Parkinson, 1980);
- Garman and Klass's, $GK = \frac{1}{2}R^2 - (2 \ln 2 - 1)c^2$ (Garman and Klass, 1980);
- Rogers and Satchell's, $RS = u(u - c) + d(d - c)$ (Rogers and Satchell, 1991);
- the daily form of Yang and Zhang's, $YZ = o^2 + k c^2 + (1 - k)RS$, with $k = 0.1393$ for a 21-session window (Yang and Zhang, 2000).

Let $T_1(\rho) = \{(1 + \rho^2) \arctan \rho - \rho\}/\pi$.

Proposition 2 (Biases under a noisy open).

- (a) For any joint law with finite second moments, $o^2 + c^2 - r^2 = -2oc$, and hence $\mathbb{E}[o^2 + c^2] - \mathbb{E}[r^2] = 2(1 - b)\mathbb{E}[o^2]$. No assumption about the error is needed.

Under Assumption 2, with η independent of (o^*, W) and Gaussian:

- (b) Close-to-close is unbiased: $\mathbb{E}[r^2] = \mathbb{E}[o^{*2}] + \sigma_c^2$. The overnight square and the open-to-close square each carry the whole error: $\mathbb{E}[o^2] - \mathbb{E}[o^{*2}] = \mathbb{E}[c^2] - \sigma_c^2 = \sigma_\eta^2$.
- (c) Parkinson. $\mathbb{E}[R^2] = \sigma_c^2\{4 \ln 2 + \Delta_R(\rho)\}$, where

$$\Delta_R(\rho) = 2T_1(\rho) + \frac{4}{\pi}(\rho - \arctan \rho) + 4 \int_0^\infty J(s) \bar{\Phi}(s/\rho) ds, \quad (4)$$

with $J(s) = \mathbb{E}[(-m)\mathbf{1}\{M < s\}]$, a one-dimensional integral of the explicit series (8) in Appendix B. Moreover, $0 \leq \Delta_R(\rho) \leq (8/\pi)\rho + \rho^2$. As $\rho \rightarrow 0$, $\Delta_R(\rho) = 2 \ln 2 \rho^2\{1 + o(1)\}$.

- (d) Rogers–Satchell. $\mathbb{E}[RS] - \sigma_c^2 = \sigma_c^2\{\rho^2/2 - T_1(\rho)\}$, exactly.
- (e) Garman–Klass. $\mathbb{E}[GK] - \sigma_c^2 = \sigma_c^2\{\Delta_R(\rho)/2 - (2 \ln 2 - 1)\rho^2\}$.
- (f) The probability that the open is the session's high or low is $(2/\pi) \arctan \rho$.
- (g) As $\sigma_\eta \rightarrow 0$ with σ_c fixed, each estimator's bias is $\gamma\sigma_\eta^2\{1 + o(1)\}$. The bias is measured against what the estimator measures when the open is exact: σ_c^2 for c^2 , Parkinson, Garman–Klass and Rogers–Satchell, and $\mathbb{E}[o^{*2}] + \sigma_c^2$ for the others. The loading γ is:

- 0 for close-to-close;
- 1 for c^2 ;
- $\frac{1}{2}$ for Parkinson and for Rogers–Satchell;
- $1 - \ln 2 = 0.307$ for Garman–Klass;
- $\frac{3}{2}$ for $o^2 + P$;
- $2 - \ln 2 = 1.307$ for $o^2 + GK$;
- $(3 + k)/2 = 1.570$ for the daily Yang–Zhang form.

Part (a) is the Yang–Zhang excess of Section 6.7 of the paper written in terms of b . The overnight and open-to-close squares together double-count by exactly twice the reversal. Part (c) is not a first-order approximation in disguise:

- the exact $\Delta_R(\rho)/(2 \ln 2 \rho^2)$ is 0.9999 at $\rho = 0.5$, 0.9936 at $\rho = 1$ and 0.9546 at $\rho = 2$;
- Parkinson therefore absorbs about half the error’s variance until the error is as large as the session’s own volatility.

Part (f) quantifies the channel by which Parkinson, which never reads the open directly, inherits the open’s error. The loadings in (g) rank the estimators by their exposure to the open. Only close-to-close has none. Garman–Klass offsets part of the range’s excess with the open-to-close square. The estimators that add the whole overnight move carry one and a half to 1.570 times the error’s variance.

Application (plan M19, part B2). Proposition 2(f) can be inverted. Under Assumption 2, the share s of non-stale opens that are the session’s high or low implies $\sigma_\eta/\sigma_c = \tan(\pi s/2)$. Under independence, b implies $\sqrt{x/(1-x)}$, where $x = (1-b)\mathbb{E}[o^2]/\mathbb{E}[c^2]$ by Proposition 1(d). In the first regime, 36.3% of non-stale opens are an extreme. The two estimates of σ_η/σ_c are:

- from that share, 0.64 [0.56, 0.72];
- from b , 0.43 [0.38, 0.50].

After the reform the share is 51.5%, and the two estimates are 1.05 and 0.96. The share-based scale is the larger in every regime, as M19 said it would be if the open is an extreme more often than an instantly corrected Gaussian error explains. A thin book makes the first trade the session’s extreme even without an error, and a slowly corrected error keeps the price near the open for longer. Both readings agree on the direction of the reform: the error grew relative to the session.

1.4 What a price band does to the unbiasedness coefficient

Proposition 3 (Censoring by a band). *Under Assumption 3:*

- $b = b_L F$, with $F = \mathbb{E}[g(o_L)o_L]/\mathbb{E}[g(o_L)^2] \geq 1$. Equality holds if and only if $P(|o_L| > \beta) = 0$.
- On interior opens ($|o_L| < \beta$), the coefficient equals b_L . On opens pinned at the band, it equals $b_L \mathbb{E}[|o_L| \mid |o_L| \geq \beta]/\beta$, which is at least b_L when $b_L \geq 0$.
- If $o_L \sim N(0, \sigma^2)$ and $k = \beta/\sigma$, then

$$F(k) = \frac{2\Phi(k) - 1}{2\Phi(k) - 1 - 2k\phi(k) + 2k^2\bar{\Phi}(k)}. \quad (5)$$

F is strictly decreasing, with $kF(k) \rightarrow 2\phi(0)$ as $k \rightarrow 0$ and $F(k) \rightarrow 1$ as $k \rightarrow \infty$. The pinned-to-interior ratio is $\lambda(k)/k$, where $\lambda = \phi/\bar{\Phi}$ is the inverse Mills ratio.

Table 2: The censoring factor of Proposition 3 against NEPSE’s regimes. The pinned share is among non-stale opens. The ratios are arithmetic on plan M15’s Table 89.

Regime	Pinned share	k	$F(k)$	Observed b/b^{int}	$\lambda(k)/k$	Observed $b^{\text{pin}}/b^{\text{int}}$
A1	0.302	1.03	1.30	1.21	1.50	1.38
B	0.226	1.21	1.21	1.15	1.40	1.29
A2	0.241	1.17	1.22	1.36	1.42	1.72
C	0.087	1.71	1.07	0.52	1.24	0.23

(d) For two regimes 0 and 1, $\ln(b_0/b_1) = \ln(b_0/b_0^{\text{int}}) + \ln(b_0^{\text{int}}/b_1^{\text{int}}) + \ln(b_1^{\text{int}}/b_1)$. Under Assumption 3 in both regimes:

- the first term is $\ln F_0 \geq 0$, strictly positive when the band binds (censoring);
- the second is the change in the latent coefficient;
- the third is $-\ln F_1 \leq 0$.

A positive third term contradicts a linear $\mathbb{E}[r \mid o_L]$ in regime 1.

The numerator in (5) is Stein’s identity: $\mathbb{E}[g(z)z] = \mathbb{E}[g'(z)]$ is the probability of an interior open. A tight band raises the measured coefficient mechanically, because a pinned open understates the move the close completes.

Application (plan M19, part B3). Table 2 sets the Gaussian factor, computed from each regime’s pinned share, against the ratio the data show. Under the $\pm 2\%$ band:

- between 0.226 and 0.302 of non-stale opens were pinned, which puts the band 1.03–1.21 standard deviations out;
- the factor is 1.21–1.30;
- the observed b/b^{int} is 1.21 in A1, 1.15 in B and 1.36 in A2.

The observed ratios fall on both sides of the Gaussian values. The censoring factor and the pinned-to-interior ratio are below them in A1 and B, and above them in A2. The Gaussian law therefore gives the order of magnitude of the band’s effect, not its exact size.

Part (d) decomposes the fall of b at the band reform, from 0.356 in A2 to 0.130 in C (a factor of 2.73), into three terms:

- the end of censoring, 0.311 or 31% of the log fall;
- a change in the interior coefficient, 0.044 or 4%: the interior coefficient was 0.261 before and 0.249 after;
- a third term of 0.649, or 65%.

The third term is positive. By (d) that is impossible if $\mathbb{E}[r \mid o_L]$ is linear in C. Opens that the old band would have pinned were reversed more than interior ones after the widening ($b = 0.130$ overall against 0.249 inside $\pm 1.9\%$). This is the overshoot recorded in Section 6.7. On this decomposition, about a third of the fall in b is the band’s mechanical inflation coming to an end. Almost two thirds is the overshoot of the large opens the band used to cap.

1.5 One coefficient pooled across securities whose opens differ

Securities i are forecast with f_{it} for a target Y_{it} , the mean squared close-to-close return over the next h sessions. The QLIKE loss is $\text{QL}(Y, f) = Y/f - \ln(Y/f) - 1$, which ranks forecasts as the latent variance would (Patton, 2011). Write $c_i^* = \mathbb{E}_i[Y/f]$, the mean over security i ’s forecasts, and $q(c) = c - 1 - \ln c \geq 0$.

Proposition 4 (Pooling).

- (a) *The shape-plus-level identity, for every security: $\mathbb{E}_i[\text{QL}(Y, f)] = \mathbb{E}_i[\text{QL}(Y, c_i^* f)] + q(c_i^*)$. The first term is the security's shape loss, the loss left after the best rescaling of its forecasts. The second is its level loss. The identity holds exactly for sample means.*
- (b) *A common rescaling κ of all forecasts that minimises the pooled loss sets the row-weighted mean of the c_i^* to one. The pooled level loss is then $\sum_i w_i q(c_i^*) \approx \frac{1}{2} \sum_i w_i (\ln c_i^*)^2$, the cross-sectional dispersion of the securities' calibration.*
- (c) *Suppose Assumptions 1–2 hold for each security, with Gaussian errors independent of the efficient returns. For the kernel $K(\beta) = (\beta o)^2 + R^2/(4 \ln 2)$, the level ratio $\psi_i(\beta) = \mathbb{E}[K_i(\beta)]/\mathbb{E}[r_i^2]$ is $1 - (1 - \beta^2)s_i + \beta^2 n_i + \sigma_{c,i}^2 \Delta_R(\rho_i)/(4 \ln 2 \mathbb{E}[r_i^2])$, where $s_i = \mathbb{E}[o_{i,t}^{*2}]/\mathbb{E}[r_i^2]$ and $n_i = \sigma_{\eta,i}^2/\mathbb{E}[r_i^2]$. To first order in the error, $\psi_i(\beta) = 1 - (1 - \beta^2)s_i + (\beta^2 + \frac{1}{2})n_i$.*
- (d) *In that first-order form, two cases follow:*
 - *If securities differ only in the error ($s_i \equiv s < 2/3$), the cross-sectional variance of $\ln \psi_i(\beta)$ is strictly increasing in $\beta \in [0, 1]$.*
 - *If they differ only in the overnight share, it is strictly decreasing and is zero at $\beta = 1$.*

Proposition 4 explains when reading the open costs more than it gives.

- **The cost is in the level.** A pooled calibration cannot fit every security, and reading the open makes each security's level depend on its own error variance. The level loss in (b) therefore grows with β when opens differ.
- **The gain is in the shape.** An informative open tracks the overnight variance.

In 120 simulated securities sharing one error variance, the full form had the lower loss (difference -0.0165). When the error variances were spread over two orders of magnitude, the open-free form had the lower loss (0.0102). All of that advantage was a level advantage: the level difference was 0.0161 and the shape difference -0.0059 . The security-level loss difference fell with the security's own b (Spearman -0.20 , against -0.04 with a common error).

The test (plan M19, part C). Plan M19 fixed a prediction before any of these statistics was computed on data: the open-free form gains most where a security's own open is noisiest. It also fixed the verdicts. The test uses the frozen forecasts of the five panel test spans at both horizons, 10 cases in all (Table 3). In each case the per-security loss difference d_i (full minus open-free) is correlated with the security's own b_i over the training span.

- **The level component.** The open-free form had the lower loss in 10 of the 10 cases, and its level loss was lower in 10 of them. The level component of part (b) favours the open-free form in every panel.
- **The mechanism, P4a.** The frozen reading required more: that the full form keep the better shape. That held in only 2 cases. In 8 cases the full form's shape loss was higher too, so the mechanism verdict is *not supported*. The open-free form's advantage is therefore not only a matter of calibration: reading a noisy open also worsens how well the forecast tracks.
- **The cross-section, P4b.** The prediction was confirmed in 4 cases: NEPSE and Dhaka 2023–2026 at both horizons. It was not detected in 5: Vietnam and Morocco at both horizons, and Dhaka 2009–2021 at twenty-one sessions. It was reversed in 1: Dhaka 2009–2021 at five sessions. The frozen overall verdict is therefore *not supported*. The ten cases are reported without a multiplicity adjustment.

Table 3: The pooling test of plan M19 (part C). Δ : full form minus open-free form, row-weighted mean QLIKE, split by Proposition 4(a). ρ_S : Spearman correlation of the security-level loss difference with the security’s own b (training span), with a 95% bootstrap interval over securities (1,999 resamples). Predicted: $\rho_S < 0$.

Panel	h	Securities	Δ total	Δ level	Δ shape	ρ_S [95% interval]	Verdict
NEPSE A2+C	5	247	0.0052	0.0023	0.0029	-0.24 [-0.35, -0.11]	confirmed
NEPSE A2+C	21	247	0.0044	0.0027	0.0017	-0.19 [-0.31, -0.06]	confirmed
Dhaka 2023–2026	5	351	0.0043	0.0021	0.0021	-0.25 [-0.35, -0.15]	confirmed
Dhaka 2023–2026	21	351	0.0002	0.0033	-0.0031	-0.23 [-0.33, -0.13]	confirmed
Dhaka 2009–2021	5	276	0.0036	0.0016	0.0020	0.18 [0.06, 0.30]	reversed
Dhaka 2009–2021	21	276	0.0016	0.0019	-0.0002	0.02 [-0.10, 0.14]	not detected
Vietnam 2007–2020	5	627	0.0076	0.0040	0.0036	0.00 [-0.08, 0.08]	not detected
Vietnam 2007–2020	21	631	0.0080	0.0047	0.0033	0.05 [-0.03, 0.12]	not detected
Morocco 2012–2026	5	54	0.0073	0.0030	0.0043	0.26 [-0.04, 0.52]	not detected
Morocco 2012–2026	21	54	0.0042	0.0038	0.0003	0.25 [-0.05, 0.53]	not detected

The prediction held in the two panels with the lowest pooled coefficient on their training spans, 0.28 and 0.37. It failed where the coefficient is between 0.55 and 0.71. That pattern was not predicted. It is reported here and has not been tested.

1.6 A rolling calibration after a rule change

Let $R_s = \sum_i r_{is}^2$ and $A_s = \sum_i A_{is}$ be date- s sums of squared returns and of an estimator’s kernel, and let $\kappa_t = \sum_{s \in W_t} R_s / \sum_{s \in W_t} A_s$ over the last L dates. Suppose that before a rule change at τ the date sums satisfy $R_s/N - \rho_0 a_s \rightarrow 0$ and $A_s/N - a_s \rightarrow 0$ in probability as the number of securities N grows, for a kernel mass $a_s > 0$ per security. After the change the same holds with ρ_1 .

Proposition 5 (Calibration lag).

- (a) As $N \rightarrow \infty$, $\kappa_t - \{\omega_t \rho_1 + (1 - \omega_t) \rho_0\} \rightarrow 0$ in probability, where $\omega_t = \sum_{s \in W_t, s \geq \tau} a_s / \sum_{s \in W_t} a_s$ is the new regime’s share of the window’s kernel mass. The relative bias $(1 - \omega_t)(\rho_0/\rho_1 - 1)$ vanishes only once the window has passed the change.
- (b) If the kernel mass is constant across the change, $\omega_{\tau+j} = \min\{(j+1)/L, 1\}$: the bias decays linearly over L dates.
- (c) Suppose changes of log size $\pm \delta$ arrive at rate λ per date, and within a regime $\ln \kappa_t$ has variance v/L . Then, to second order in δ and for small λL , the average level loss $\frac{1}{2} \mathbb{E}[(\ln \kappa_t - \ln \rho_t)^2]$ is $\frac{1}{2}(\lambda \delta^2 L/3 + v/L)$. It is minimised at $L^* = \sqrt{3v/(\lambda \delta^2)}$, where it equals $\sqrt{\lambda \delta^2 v/3}$.

Part (c) is the bias–variance choice of an estimation window under breaks (Pesaran and Timmermann, 2007), applied to a calibration ratio. The factor $1/3$ is $\int_0^1 (1-u)^2 du$, from the linear decay in (b).

Application (plan M19, part B4). The frozen calibration of Anam’s estimator on NEPSE is the ratio over the 60 dates before the band reform, $\rho_0 = 0.970$. Regime C’s own ratio is $\rho_1 = 0.670$. Over C’s 90 sessions, the path of Proposition 5(a) explains 99% of the squared deviation of the applied calibration from ρ_1 (root mean squared error 0.017). The linear weights of part (b) explain a share of 0.98.

The before-reform regimes give $v = 0.16$. NEPSE’s three rule boundaries changed the calibration by $\delta = -0.103, 0.156$ and -0.299 over 567 sessions. With these, part (c) puts the balancing window at $L^* \approx 46$ dates, shorter than the frozen 60.

Regime C lies wholly after a change. There, the lag alone, with a constant post-reform ratio, predicts that a 60-date calibration loses 0.002 more than a 10-date one. The observed QLIKE differences are 0.047 at five sessions and 0.053 at twenty-one, so the lag accounts for about 5% of the gap. Proposition 5(a) describes the applied calibration’s path almost exactly. The forecast loss of a long window in C, however, comes mostly from something the proposition holds fixed. The ratio itself moved within C, because the estimator’s own pooled b was still falling and so the kernel was changing, and a short window tracks that movement. The balancing window L^* assumes the same constant ratios, and is likely too long for the same reason.

1.7 Identifying an estimator’s calibration without a benchmark

Plan M14 writes each daily estimator as $X_{k,it} = \alpha_{k,i} + \beta_k IV_{it} + U_{k,it}$, where $\mathbb{E}[U_{it} \mid IV_{it}, \mathcal{F}_{i,t-1}] = 0$. The reference is $X_0 = c^2$, with $\beta_0 = 1$. Let $V_{it} = \mathbb{E}[IV_{it} \mid \mathcal{F}_{i,t-1}]$.

Proposition 6 (Instrumented calibration). *Under this model, for any $\mathcal{F}_{i,t-1}$ -measurable instruments Z with finite second moments:*

- (a) $\text{Cov}(X_k, Z) = \beta_k \text{Cov}(X_0, Z)$ for every k . The cross-covariance matrix has rank one, and $\beta_k = \text{Cov}(X_k, Z'w) / \text{Cov}(X_0, Z'w)$ whenever $\text{Cov}(V, Z'w) \neq 0$.
- (b) With IV uncorrelated with U , the least-squares slope of X_k on X_0 is $\{\beta_k \text{Var}(IV) + \text{Cov}(U_k, U_0)\} / \{\text{Var}(IV) + \text{Var}(U_0)\}$. The mean ratio is $\beta_k + \delta_k$, with $\delta_k = (\mathbb{E}X_k - \beta_k \mathbb{E}X_0) / \mathbb{E}X_0$. Neither is β_k in general.
- (c) If an instrument dated $t - \ell$ is correlated with the measurement errors, the instrumented estimand is $\{\beta_k \text{Cov}(V, Z) + \text{Cov}(U_k, Z)\} / \{\text{Cov}(V, Z) + \text{Cov}(U_0, Z)\}$. If the errors are moving averages of order q with respect to $\mathcal{F}$, instruments dated $t - q - 1$ or earlier are valid.
- (d) The two-stage least-squares estimator with forward orthogonal deviations is consistent and asymptotically normal under standard mixing and moment conditions. Hansen’s statistic is asymptotically χ^2 with one degree of freedom fewer than the number of excluded instruments (Hansen, 1982; Arellano and Bover, 1995). With moving-average errors, its covariance must allow for serial correlation of the same order.
- (e) Among composites $w'X$ with $w'\beta = 1$, the variance is minimised by $w^* = \Sigma_X^{-1}\beta / (\beta'\Sigma_X^{-1}\beta)$. This equals $\Omega_U^{-1}\beta / (\beta'\Omega_U^{-1}\beta)$ when $\Sigma_X = \text{Var}(IV)\beta\beta' + \Omega_U$.

Part (c) qualifies the identification behind Section 6.6.

- **Valid instruments.** In 100 simulated panels with serially independent errors, instruments dated $t - 1$ recovered slopes of 0.8 and 1.3 as 0.794 and 1.295. The J test rejected at 5% in a share of 0.09 of the panels.
- **A persistent error.** With first-order moving-average errors, such as an opening error that persists into the next day’s bar, the same instruments gave 0.572 and 0.522, and J rejected in a share of 1.00.
- **Deeper lags.** Instruments dated $t - 2$ restored 0.789 and 1.294.

The two-session-lag sensitivity reported in Section 6.6 is therefore the relevant check of identification, not an optional one.

1.8 What Anam’s kernel does with a noisy open

An independent audit of 9 October 2026 objected to two readings of Anam’s estimator (Section 6.8 of the paper). The first is that bo is the permanent overnight move, and its square that move’s variance. The second is that extending the range removes an open that overshoots. This

subsection replaces both readings with a result, and measures how much of the opening error each kernel absorbs. It was derived after the audit, and nothing in it was predicted in advance.

Measured from the previous close, write $h = \ln(H/C_-)$ and $l = \ln(L/C_-)$. The kernel of Anam's estimator, with $w = \lambda_0(1 - b)$ and $\lambda_0 = 0.2$, is

$$A = (1 - w) \left\{ (bo)^2 + \frac{R^{*2}}{4 \ln 2} \right\} + w r^2, \quad R^* = R + \max(0, bo - h) + \max(0, l - bo). \quad (6)$$

Its open-free form is $A_0 = (1 - \lambda_0) TR^2 / (4 \ln 2) + \lambda_0 r^2$, where $TR = \max(h, 0) - \min(l, 0)$ is Wilder's true range (Wilder, 1978). It is the case $b = 0$. Let $\pi(o) = b^2 o^2 + b(1 - b) \mathbb{E}[o^2]$, and let $\psi(a) = \mathbb{E}[(Z - a)^+] = \phi(a) - a \bar{\Phi}(a)$ for a standard normal Z .

Proposition 7 (Anam's kernel under a noisy open).

(a) Under Assumption 1, $\mathbb{E}[(bo)^2] = b^2 \mathbb{E}[o^2] = \mathbb{E}[o^{*2}] - \mathbb{E}[(o^* - bo)^2] \leq \mathbb{E}[o^{*2}]$, with equality if and only if $o^* = bo$ almost surely.

- If $\mathbb{E}[o^* \eta] = 0$, then $\mathbb{E}[(bo)^2] = b \mathbb{E}[o^{*2}]$, and $\mathbb{E}[\pi(o)] = \mathbb{E}[o^{*2}]$.
- If, in addition, (o^*, η) is a centred Gaussian vector, then $\pi(o) = \mathbb{E}[o^{*2} | o]$.
- Without a restriction on $\mathbb{E}[o^* \eta]$, the deficit $\mathbb{E}[(o^* - bo)^2]$ is identified only up to the interval $[0, \mathbb{E}[r^2] - b^2 \mathbb{E}[o^2]]$.

(b) For every bar, $R^* = \max(h, bo) - \min(l, bo)$. Hence $R \leq R^* \leq TR$ for $b \in [0, 1]$, with $R^* = R$ when $b = 1$, so the extension never removes a printed open that set the high or the low. With $C_- = C = L$ and $O = H$, the open-free kernel is $A_0 = (1 - \lambda_0) \ln^2(O/C_-) / (4 \ln 2) > 0$, although nothing but the print moved.

(c) Let Assumption 2 hold, with $o^* \sim N(0, s^2 \sigma_c^2)$ independent of (η, W) and b at its population value $s^2 / (s^2 + \rho^2)$. As $\sigma_\eta \rightarrow 0$ with σ_c and s fixed, each kernel's expectation exceeds its value at $\sigma_\eta = 0$ by $\gamma \sigma_\eta^2 \{1 + o(1)\}$. The loadings γ are:

- for the true range, $TR^2 / (4 \ln 2)$,

$$\gamma_{TR}(s) = \frac{1}{4} + \frac{s \mathbb{E}[\psi(U/s)]}{2\sqrt{2\pi} \ln 2}. \quad (7)$$

Here U is the maximum of a Brownian meander on $[0, 1]$, with $P(U \leq u) = 1 + 2 \sum_{k \geq 1} (-1)^k e^{-k^2 u^2 / 2}$. The loading γ_{TR} increases in s , from $\frac{1}{4}$ as $s \rightarrow 0$ to $s / (4\pi \ln 2) + o(1)$ as $s \rightarrow \infty$, and $\max\{\frac{1}{4}, s / (4\pi \ln 2)\} \leq \gamma_{TR}(s) \leq \frac{1}{4} + s / (4\pi \ln 2)$;

- $(1 - \lambda_0) \gamma_{TR}(s)$ for the open-free kernel A_0 ;
- $-\frac{1}{2}$ for the full kernel A , whatever s ;
- $+\frac{1}{2}$ for A with $\pi(o)$ in place of $(bo)^2$.

Part (a) makes the audit's first objection exact. The overnight term is the lower end of the identified set of Proposition 1(e). It is unbiased for the efficient overnight second moment only under proportional overreaction. Under an error uncorrelated with the news it carries the fraction b of that moment. In NEPSE, where b lies between 0.13 and 0.36, that is a third or less. The calibration κ restores the average level of the whole kernel, not the overnight share within it. The posterior moment $\pi(o)$ corrects the share, but only under the uncorrelated-error reading, which the data cannot confirm.

Part (b) makes the second objection exact. The extension moves the anchor; it does not move the extremes. The audit's bar, $C_- = C = L = 100$ and $O = H = 105$, gives $A_0 = 6.87 \times 10^{-4}$, all of it opening error.

Part (c) measures what remains:

Table 4: Each kernel’s exposure to the opening error: the excess of its expectation over its value with an exact open, in units of σ_η^2 (Assumption 2, $o^* \sim N(0, s^2\sigma_c^2)$, b at its population value, $\sigma_c = 1$). “First order” is Proposition 7(c) and Proposition 2(g). The other columns are exact, by simulation with 4,000,000 paths. Ledger: `table134_theory_kernel_exposure.csv`.

Kernel	$s = 0.3$			$s = 1$		
	First order	$\rho = 0.5$	$\rho = 1$	First order	$\rho = 0.5$	$\rho = 1$
close-to-close r^2	0.000	0.000	0.000	0.000	0.000	0.000
Parkinson $R^2/(4 \ln 2)$	0.500	0.500	0.497	0.500	0.500	0.497
true range $TR^2/(4 \ln 2)$	0.250	0.451	0.481	0.258	0.366	0.430
open-free kernel A_0	0.200	0.361	0.385	0.206	0.292	0.344
full kernel A	−0.500	0.243	0.353	−0.500	−0.205	0.069
A with $\pi(o)$ for $(bo)^2$	0.500	0.469	0.421	0.500	0.563	0.519
$o^2 + P$	1.500	1.501	1.498	1.500	1.501	1.498

- **To first order**, the open-free kernel absorbs $(1 - \lambda_0)\gamma_{TR}(s)$ of the error’s variance: 0.200 at $s = 0.3$ and 0.206 at $s = 1$. Parkinson absorbs $\frac{1}{2}$ and overnight² + Parkinson $\frac{3}{2}$ (Proposition 2(g)).
- **The full kernel over-corrects**. Its shrinkage of the overnight move removes the residual uncertainty as well as the error, by part (a). To first order this outweighs the half that the range absorbs.
- **First order needs $\sigma_\eta \ll s\sigma_c$** . At the error sizes of NEPSE this condition fails, so Table 4 gives the exact exposures, by simulation. Under an uncorrelated error, the moments of NEPSE’s first regime imply $s = 0.29$ and $\rho = 0.43$ (and $s = 0.37$, $\rho = 0.96$ after the reform). At $s = 0.3$ and $\rho = 0.5$ the open-free kernel absorbs 0.361 of the error’s variance, against 0.500 for Parkinson and 0.243 for the full kernel.
- **Across the samples**. At each sample’s own implied s and ρ (nine samples: NEPSE’s four regimes, the NIFTY 50 and the four frontier panels), the open-free kernel absorbs between 0.25 and 0.38 of the error’s variance and Parkinson between 0.50 and 0.50. The full kernel ranges from −0.36 to 0.33: its overnight shrinkage over-corrects where the overnight move is large relative to the session, and under-corrects where it is small. These are reported, not tested: they rest on the uncorrelated-error reading of the moments.

Leaving the open out therefore reduces the exposure to the opening error but does not remove it. Which of the two forms is exposed more depends on s and ρ . Only close-to-close has no exposure at all.

1.9 The market-implied open

Propositions 1 and 7 treat one security’s open. A panel offers more. On each date many securities open together. Each opening error is the security’s own, while part of each efficient overnight move is common to all. This subsection states what the cross-section can recover and what it cannot. It was derived during the development of Anam II, the estimator of plan M22, and was written after that plan was frozen.

Assumption 4 (A common overnight factor, and errors that are each security’s own). On one date, securities $i = 1, \dots, N$ have $o_i^* = \beta_i f + \zeta_i$, $o_i = o_i^* + \eta_i$ and $r_i = o_i^* + x_i$. Here:

- f is the market’s efficient overnight move, with variance σ_f^2 ;
- the pairs (ζ_i, η_i) have zero means and finite second moments, and are independent across securities and of f ;

- within a pair, a security's error η_i may be correlated with its own news ζ_i ;
- nothing known at the open, including the other securities' opens, predicts the rest of any session: $\mathbb{E}[x_i \mid f, (\zeta_j, \eta_j)_{j \leq N}] = 0$.

The leave-one-out mean of the overnight returns, $m_{-i} = (N-1)^{-1} \sum_{j \neq i} o_j$, is the *market-implied open* of security i . Write:

- $\bar{\beta}_{-i}$ for the mean of the other securities' β_j ;
- $V_{N,i} = (N-1)^{-2} \sum_{j \neq i} \mathbb{E}[(\zeta_j + \eta_j)^2]$;
- $\mu = \sum_i \beta_i^2 \sigma_f^2 / \sum_i \mathbb{E}[o_i^2]$, the market's share of the overnight second moment;
- $b_I = \sum_i \{\mathbb{E}[\zeta_i^2] + \mathbb{E}[\zeta_i \eta_i]\} / \sum_i \mathbb{E}[(\zeta_i + \eta_i)^2]$, the pooled reliability of the securities' own parts.

Proposition 8 (The market-implied open). *Under Assumption 4:*

- m_{-i} is a function of f and of the other securities' (ζ_j, η_j) only. It is therefore independent of security i 's own news and error (ζ_i, η_i) , and it does not move when security i 's opening print does.
- $\mathbb{E}[(m_{-i} - \bar{\beta}_{-i} f)^2] = V_{N,i} \leq \max_j \mathbb{E}[(\zeta_j + \eta_j)^2] / (N-1)$. The market-implied open converges in mean square to the market's efficient overnight move as the cross-section grows.
- $\mathbb{E}[m_{-i}^2] = \bar{\beta}_{-i}^2 \sigma_f^2 + V_{N,i}$, and the pooled projection of the close-to-close return on the market-implied open is

$$b_M = \frac{\sum_i \beta_i \bar{\beta}_{-i} \sigma_f^2}{\sum_i (\bar{\beta}_{-i}^2 \sigma_f^2 + V_{N,i})}.$$

With equal betas it is $\beta^2 \sigma_f^2 / (\beta^2 \sigma_f^2 + V_N)$. This tends to one: the session keeps the market's move. A finite cross-section attenuates it.

- The coefficient of Proposition 1, pooled over the securities, is

$$b = \mu \cdot 1 + (1 - \mu) b_I,$$

the mean of the two parts' reliabilities weighted by their shares of the overnight second moment. In a sample the same decomposition is an identity. Take o_M as the date's cross-sectional mean, including i , and $o_I = o - o_M$. Then $\hat{b} = \hat{\mu} \hat{b}_M + (1 - \hat{\mu}) \hat{b}_I$ exactly, with $\hat{b}_M$, $\hat{b}_I$ and $\hat{\mu}$ the sample counterparts.

- The kernel of Anam II is

$$A = (1 - w) \left\{ o^{*2} + \frac{R^{*2}}{D} \right\} + w r^2, \quad R^* = \max(h, o^*) - \min(l, o^*).$$

Here $o^* = m_{-i}$ when the open printed at the previous close ($o_i = 0$), and $o^* = 0$ otherwise. The divisor is $D = 1$ on a one-price bar and $D = 4 \ln 2$ otherwise. Two cases follow:

- on a bar whose open moved and which has a range, A is the open-free kernel of Proposition 7;
 - on a bar that never left the previous close, $A = 2(1 - w) m_{-i}^2$, with expectation $2(1 - w)(\bar{\beta}_{-i}^2 \sigma_f^2 + V_{N,i})$, while the open-free kernel is zero.
- Let $\lambda_n = \mathbb{E}[R_n^2]$, where $R_n = \max\{0, W(\frac{1}{n}), \dots, W(1)\} - \min\{0, W(\frac{1}{n}), \dots, W(1)\}$ for a standard Brownian motion observed n times after its start. Then:

- $\lambda_1 = 1$;

Table 5: The open’s reliability by component, on each panel’s training span (Proposition 8). o_M is the date’s cross-sectional mean overnight return, $o_I = o - o_M$, and $\hat{\mu} = \sum o_M^2 / \sum o^2$. “LOO” is the pooled projection of r on the leave-one-out mean m_{-i} ; “predicted” is part (c)’s attenuation, $1 - \sum V_N / \sum m^2$, with V_N the date’s cross-sectional variance of o over $N - 1$. [†]Pakistan entered plan M22 unseen; it is reported here after that plan’s evaluation. Ledger: `table144_theory_market_open_data.csv`.

Panel	N/date	$\hat{b}$	$\hat{b}_M$	$\hat{b}_I$	$\hat{\mu}$	LOO b_M	Predicted	Stale opens	One price
NEPSE	244	0.279	0.907	0.200	11.3%	0.900	0.968	8.2%	0.1%
Dhaka 2023–2026	332	0.371	0.635	0.326	14.5%	0.629	0.982	37.2%	24.5%
Dhaka 2009–2021	223	0.550	0.954	0.446	20.5%	0.946	0.983	7.2%	1.6%
Vietnam 2007–2020	467	0.592	1.004	0.515	15.7%	0.997	0.986	22.2%	13.7%
Morocco 2012–2026	30	0.707	0.779	0.702	5.9%	0.373	0.425	19.9%	26.7%
Pakistan 2016–2026 [†]	87	0.582	1.093	0.470	17.9%	1.067	0.946	5.5%	2.5%

- $\lambda_n \leq \lambda_{kn}$ for every integer $k \geq 1$;
- $\lambda_n \leq 4 \ln 2$, and $\lambda_n \rightarrow 4 \ln 2$.

A bar with one price is the case $n = 1$, so its divisor is 1, not Parkinson’s $4 \ln 2$.

Why a single coefficient failed. Part (d) is why reading the open through one coefficient did not beat ignoring it in plan M20. On the training spans of the five frontier panels:

- the session keeps the market part of the overnight move almost entirely, with $\hat{b}_M$ between 0.63 and 1.00;
- it reverses most of the securities’ own parts, with $\hat{b}_I$ between 0.20 and 0.70;
- the market part is only 6–20% of the overnight second moment.

So $\hat{b}$ is close to $\hat{b}_I$ (Table 5). One coefficient shrinks the reliable component by the unreliable one’s factor.

The market’s move is measured without the stock’s error. Parts (a)–(c) say how far it can be measured. A large cross-section measures it almost exactly. Morocco’s 30 securities a date attenuate the leave-one-out projection. Part (c) predicts that attenuation from each date’s cross-sectional dispersion, as $1 - \sum V_N / \sum m^2$ (the table’s last column but two).

Stale bars and stale opens. A stale print is the case of non-synchronous trading that Scholes and Williams (1977) and Lo and MacKinlay (1990) studied for returns. A security that did not trade away from the previous close has a latent price that still moved with the market, and its later returns realise that move. Part (e) shows how the kernel credits it. The factor 2 counts both the move and the step back to the stale print; the calibration κ sets the level.

Why only stale opens. Anam II reads the market’s move only where the open printed at the previous close, because the move is common only on average. A security with $\beta_i \neq \bar{\beta}$ would be credited $\bar{\beta}^2 \sigma_f^2$ in place of $\beta_i^2 \sigma_f^2$. By Proposition 4(b), such a misallocation that differs across securities is a level loss under a pooled calibration. The development record shows this cost where betas are dispersed, in Vietnam.

The one-price divisor. Part (f) is a fact about discrete monitoring (Broadie et al., 1997; Christensen and Podolskij, 2007). Parkinson’s constant is the limit of many observations. Simulation puts λ_n at 1.00, 1.22, 1.58, 2.07 and 2.51 for $n = 1, 2, 5, 20, 200$, against $4 \ln 2 = 2.77$.

Verification. The script `54_market_open_theory.py` runs 35 checks, of which 35 pass (ledger: table 143). Across the six panels, the sample identity of part (d) holds to within 1×10^{-16} .

1.10 Verification

Every closed form, identity and inequality above was set against one of three references by `scripts/45_theory_checks.py`:

- a Monte Carlo simulation, in which Brownian extremes were drawn exactly from the Brownian bridge within each step and the closed form had to lie within four standard errors;
- a numerical integral, with a tolerance of 10^{-6} ;
- exact arithmetic, for identities and inequalities.

The ledger, `output/tables/table121_theory_checks.csv`, has 187 rows, of which 187 pass and 0 fail (Table 6).

Table 6: Checks of the propositions against simulation. Ledger: `table121_theory_checks.csv`.

Proposition	Checks	Passed
1	41	41
2	76	76
3	35	35
4	19	19
5	4	4
6	12	12

A second script, `scripts/46_theory_proofs.py`, then went through every part of every proposition and every step of every proof in Appendix B. It also covered the paper’s own mathematical claims: the Yang–Zhang identity, the non-negativity of Garman–Klass on a valid bar, the special cases of Anam’s kernel, and the old and corrected Appendix A. Each step was verified in one of five ways:

- **Symbolically** (166 steps). SymPy reduces the difference of the two sides to zero. For 3 of these steps, where SymPy builds both sides exactly but its simplifier does not close the difference, the difference was instead evaluated to 60 significant digits at twelve random points and had to vanish to 50.
- **Numerically** (66 steps), by quadrature, series or exact arithmetic on a grid. This includes monotonicity in 50-digit arithmetic.
- **By simulation** (51 steps), within four standard errors. This is how the results the proofs cite from the literature were checked where the proofs use them: the reflection principle, the method of images, the meander limit and the GMM asymptotics.
- **Pathwise** (31 steps), as identities on every simulated path.
- **On the data** (7 steps), as identities on every NEPSE stock-day they apply to.

The step ledger, `output/tables/table124_theory_proof_steps.csv`, has 321 steps, of which 321 pass and 0 fail (Table 7). Proposition 7 was added after both scripts’ first runs. Its closed forms were set against simulation and numerical integrals by `scripts/48_kernel_theory.py`, in 63 checks, of which 63 pass (`table133_theory_kernel_checks.csv`). Its proof is in the step ledger with the others.

B Proofs

All moments are raw, Φ and ϕ are the standard normal distribution and density, and $\bar{\Phi} = 1 - \Phi$. For a standard Brownian motion W on $[0, 1]$, let $M = \sup W$, $m = \inf W$ and

Table 7: Every step of every proof, by how it was verified. Ledger: `table124_theory_proof_steps.csv`.

	Symbolic	Numerical	Simulation	Pathwise	Data	Steps	Passed
The model	3	0	0	0	0	3	3
Proposition 1	30	4	0	0	1	35	35
Proposition 2	33	29	26	18	0	106	106
Proposition 3	23	13	0	0	0	36	36
Proposition 4	17	6	0	0	0	23	23
Proposition 5	11	3	0	0	0	14	14
Proposition 6	10	1	5	0	0	16	16
Proposition 7	28	9	20	13	0	70	70
The paper's own claims	11	1	0	0	6	18	18

$D = W(1)$. Each step below was verified by `scripts/46_theory_proofs.py`, symbolically wherever it is algebra or calculus (Table 7). The closed forms were also set against simulation by `scripts/45_theory_checks.py` (Table 6).

B.1 Proof of Proposition 1

(a) By Assumption 1, $\mathbb{E}[ox] = \mathbb{E}[o^*x] + \mathbb{E}[\eta x] = 0$. Since $r = o^* + x$,

$$\mathbb{E}[or] = \mathbb{E}[oo^*] + \mathbb{E}[ox] = \mathbb{E}[oo^*].$$

The map $\gamma \mapsto \mathbb{E}[(o^* - \gamma o)^2] = \mathbb{E}[o^{*2}] - 2\gamma\mathbb{E}[oo^*] + \gamma^2\mathbb{E}[o^2]$ is a convex quadratic minimised at $\gamma = \mathbb{E}[oo^*]/\mathbb{E}[o^2] = b$. Hence bo is the best linear predictor of o^* , and $C_{-e^{bo}}$ that of the efficient open on the log scale. If $\mathbb{E}[o^* | o] = \gamma o$, then $\mathbb{E}[oo^*] = \mathbb{E}[o\mathbb{E}[o^* | o]] = \gamma\mathbb{E}[o^2]$, so $\gamma = b$.

(b) Since $c = r - o$, we have $\mathbb{E}[oc] = \mathbb{E}[or] - \mathbb{E}[o^2] = (b - 1)\mathbb{E}[o^2]$. Only the definition of b is used.

(c) Write $\eta = o - o^* = (1 - b)o - (o^* - bo)$. Then

$$\mathbb{E}[\eta^2] = (1 - b)^2\mathbb{E}[o^2] - 2(1 - b)\mathbb{E}[o(o^* - bo)] + \mathbb{E}[(o^* - bo)^2].$$

The middle term vanishes, because $\mathbb{E}[o(o^* - bo)] = \mathbb{E}[oo^*] - b\mathbb{E}[o^2] = 0$ by (a). The last term is non-negative, and it is zero if and only if $o^* = bo$ almost surely. That is the stated condition. For $b > 0$, it is equivalent to $\eta = \{(1 - b)/b\}o^*$: an error proportional to the news.

(d) The three cases:

- If $\mathbb{E}[o^*\eta] = 0$, then $\mathbb{E}[oo^*] = \mathbb{E}[o^{*2}]$ and $\mathbb{E}[o^2] = \mathbb{E}[o^{*2}] + \mathbb{E}[\eta^2]$, so $(1 - b)\mathbb{E}[o^2] = \mathbb{E}[o^2] - \mathbb{E}[o^{*2}] = \mathbb{E}[\eta^2]$.
- If $\eta = \theta o^* + \nu$ with $\mathbb{E}[o^*\nu] = 0$, then $o = (1 + \theta)o^* + \nu$. Hence $\mathbb{E}[oo^*] = (1 + \theta)\mathbb{E}[o^{*2}]$ and $\mathbb{E}[o^2] = (1 + \theta)^2\mathbb{E}[o^{*2}] + \mathbb{E}[\nu^2]$.
- The stale open is the case $\theta = \lambda - 1$ and $\nu = 0$, which gives $b = 1/\lambda$.

(e) Write $m_{oo} = \mathbb{E}[o^2]$ and $m_{rr} = \mathbb{E}[r^2]$. By (a), $\mathbb{E}[or] = bm_{oo}$ and $\mathbb{E}[r^2] = \mathbb{E}[o^{*2}] + \mathbb{E}[x^2]$, since $\mathbb{E}[o^*x] = 0$.

- *Necessity.* By (c), $0 \leq \mathbb{E}[(o^* - bo)^2] = \mathbb{E}[o^{*2}] - b^2m_{oo}$, and $\mathbb{E}[o^{*2}] \leq m_{rr}$.
- *Sufficiency.* Fix $s \in [b^2m_{oo}, m_{rr}]$. Let (o, o^*) be a centred Gaussian vector with $\mathbb{E}[o^2] = m_{oo}$, $\mathbb{E}[oo^*] = bm_{oo}$ and $\mathbb{E}[o^{*2}] = s$. Its covariance matrix is positive semi-definite because $sm_{oo} \geq b^2m_{oo}^2$. Let x be an independent centred Gaussian with $\mathbb{E}[x^2] = m_{rr} - s$, and set $\eta = o - o^*$.

In the sufficiency construction, Assumption 1 holds. The vector $(o, r) = (o, o^* + x)$ is centred Gaussian with $\mathbb{E}[o^2] = m_{oo}$, $\mathbb{E}[or] = bm_{oo}$ and $\mathbb{E}[r^2] = m_{rr}$, the same law for every s . The range of $\mathbb{E}[\eta^2]$ follows from (c) with $\mathbb{E}[(o^* - bo)^2] = s - b^2m_{oo}$.

The published bound. Let $b < 1$. Then $3 - 2b > 1$, so both sides of the second inequality below are positive and squaring preserves it:

$$\frac{1}{2}\{\sqrt{5-4b}-1\} < 1-b \iff \sqrt{5-4b} < 3-2b \iff 5-4b < 9-12b+4b^2 \iff 0 < 4(1-b)^2.$$

So for every $b < 1$ the bound of Appendix A is strictly below (3), and by (c) it is never attained. $\square$

B.2 Proof of Proposition 2

(a) The identity is $(o+c)^2 = o^2 + 2oc + c^2$. Taking expectations and using Proposition 1(b) gives the result.

By Brownian scaling, we take $\sigma_c = 1$ from here on. Every expectation below scales with σ_c^2 , and $\rho = \sigma_\eta$. Measured from the efficient open, the printed path is $\{\eta\} \cup \{W(s) : s \in (0, 1]\}$. Its extremes are $\tilde{h} = \max(\eta, M)$ and $\tilde{l} = \min(\eta, m)$. Then:

$$R = \tilde{h} - \tilde{l}, \quad u = \tilde{h} - \eta, \quad d = \tilde{l} - \eta, \quad c = D - \eta, \quad r = o^* + D.$$

(b) Independence gives $\mathbb{E}[r^2] = \mathbb{E}[o^{*2}] + 1$ and $\mathbb{E}[o^2] - \mathbb{E}[o^{*2}] = \mathbb{E}[\eta^2]$. The same argument gives $\mathbb{E}[c^2] - 1 = \mathbb{E}[\eta^2] - 2\mathbb{E}[D\eta] = \mathbb{E}[\eta^2]$.

(c), *the representation.* Since $M \geq 0 \geq m$, the range has three cases:

- $R = M - m$ if $\eta \in [m, M]$;
- $R = \eta - m$ if $\eta > M$;
- $R = M - \eta$ if $\eta < m$.

Hence $R = (M - m) + (\eta - M)^+ + (m - \eta)^+$, with at most one of the last two terms positive. This gives $M - m \leq R \leq M - m + |\eta|$, because $(\eta - M)^+ \leq \eta^+$ and $(m - \eta)^+ \leq \eta^-$. Squaring,

$$R^2 - (M - m)^2 = (\eta - M)(\eta + M - 2m)\mathbf{1}\{\eta > M\} + (m - \eta)(2M - m - \eta)\mathbf{1}\{\eta < m\}.$$

The map $(W, \eta) \mapsto (-W, -\eta)$ preserves the joint law and exchanges the two terms. So

$$\Delta_R := \mathbb{E}[R^2] - 4\ln 2 = 2\mathbb{E}[(\eta - M)^+]^2 + 4\mathbb{E}[(\eta - M)^+M] + 4\mathbb{E}[(\eta - M)^+(-m)].$$

Here $\mathbb{E}[(M - m)^2] = 4\ln 2$ (Feller, 1951; Parkinson, 1980). Call the three expectations T_1 , T_2 and T_3 .

(c), T_1 and T_2 . By the reflection principle, M has the law of $|Z'|$ with Z' standard normal, independent of $\eta = \rho Z$. Write $(Z, Z') = \varrho(\cos \varphi, \sin \varphi)$, so that $\varrho^2 \sim \chi_2^2$ (with $\mathbb{E}\varrho^2 = 2$) is independent of $\varphi \sim U(-\pi, \pi]$. Then $(\rho Z - |Z'|)^+ = \varrho(\rho \cos \varphi - |\sin \varphi|)^+$. This is positive exactly when $|\varphi| < \alpha := \arctan \rho$. Hence

$$T_1 = \frac{2}{\pi} \int_0^\alpha (\rho \cos \varphi - \sin \varphi)^2 d\varphi = \frac{(1 + \rho^2)\alpha - \rho}{\pi},$$

$$T_2 = \frac{2}{\pi} \int_0^\alpha (\rho \cos \varphi - \sin \varphi) \sin \varphi d\varphi = \frac{\rho - \alpha}{\pi}.$$

Both integrals use $\sin^2 \alpha = \rho^2/(1 + \rho^2)$ and $\sin 2\alpha = 2\rho/(1 + \rho^2)$.

(c), T_3 . For $y > 0$, $(y - M)^+ = \int_0^y \mathbf{1}\{M < s\} ds$. Conditioning on η and using Fubini,

$$T_3 = \int_0^\infty J(s) P(\eta > s) ds = \int_0^\infty J(s) \bar{\Phi}(s/\rho) ds,$$

$$J(s) = \mathbb{E}[(-m)\mathbf{1}\{M < s\}] = \int_0^\infty \{P(M < s) - G(s, t)\} dt.$$

Here $P(M < s) = 2\Phi(s) - 1$. The joint distribution of the extremes is given by the method of images (e.g. Borodin and Salminen, 2002): for $a, b > 0$ and $\ell = a + b$,

$$G(a, b) = P(M < a, -m < b) \\ = \sum_{k=-\infty}^{\infty} \{\Phi(a + 2k\ell) - \Phi(-b + 2k\ell) - \Phi(a + 2b + 2k\ell) + \Phi(b + 2k\ell)\}. \quad (8)$$

Substituting T_1, T_2 and T_3 gives (4).

(c), *the bounds.* $\Delta_R \geq 0$ is immediate from $R \geq M - m$. For the upper bound, use $R \leq M - m + |\eta|$, $\mathbb{E}[M - m] = 2\sqrt{2/\pi}$ and $\mathbb{E}|\eta| = \rho\sqrt{2/\pi}$:

$$\mathbb{E}[R^2] \leq 4 \ln 2 + \frac{8}{\pi} \rho + \rho^2.$$

(c), *the small-error limit.* As $\rho \rightarrow 0$, $(1 + \rho^2) \arctan \rho - \rho = \frac{2}{3}\rho^3 + O(\rho^5)$ and $\rho - \arctan \rho = \frac{1}{3}\rho^3 + O(\rho^5)$, so T_1 and T_2 are $O(\rho^3)$. Substituting $s = \rho v$,

$$T_3 = \rho^2 \int_0^\infty \frac{J(\rho v)}{\rho v} v \bar{\Phi}(v) dv.$$

Conditional on $M < s$, the path $-W$ converges as $s \downarrow 0$ to a Brownian meander (Durrett et al., 1977). The meander's maximum has mean $\sqrt{2\pi} \ln 2$ (Durrett and Iglehart, 1977), and the maximum's density at zero is $f_M(0) = \sqrt{2/\pi}$. Hence

$$J(s)/s \rightarrow f_M(0) \sqrt{2\pi} \ln 2 = 2 \ln 2.$$

The check script confirms the limit numerically from (8).

The ratio $J(s)/s$ is bounded:

- on $(0, 1]$, because it is continuous with a finite limit at zero;
- for $s > 1$, by $\mathbb{E}[-m] = \sqrt{2/\pi}$.

Dominated convergence therefore gives $T_3/\rho^2 \rightarrow 2 \ln 2 \int_0^\infty v \bar{\Phi}(v) dv = \frac{1}{2} \ln 2$. So $\Delta_R = 4T_3 + O(\rho^3) = 2 \ln 2 \rho^2 \{1 + o(1)\}$.

(d) Write $RS = (\tilde{h} - \eta)(\tilde{h} - D) + (\tilde{l} - \eta)(\tilde{l} - D)$, using $u - c = \tilde{h} - D$ and $d - c = \tilde{l} - D$. Without an error, $RS_0 = M(M - D) + m(m - D)$, with $\mathbb{E}[RS_0] = 1$ (Rogers and Satchell, 1991). The difference $RS - RS_0$ is:

- $-\eta(M + m - 2D)$ on $\{m \leq \eta \leq M\}$;
- $-\eta(m - D) - M(M - D)$ on $\{\eta > M\}$;
- the mirror image of the second case on $\{\eta < m\}$.

Since $\mathbb{E}[\eta(M + m - 2D)] = 0$, the first case contributes $\mathbb{E}[\eta(M + m - 2D)(\mathbf{1}\{\eta > M\} + \mathbf{1}\{\eta < m\})]$. Collecting terms and using the symmetry,

$$\mathbb{E}[RS] - 1 = 2\mathbb{E}[\mathbf{1}\{\eta > M\}(\eta - M)(M - D)] = 2\mathbb{E}[(\eta - M)^+(M - D)].$$

By the reflection principle, (M, D) has density $2(2a - x)\phi(2a - x)$ on $a \geq \max(0, x)$. Substituting $z = 2a - x$ gives

$$\mathbb{E}[(M - D)\mathbf{1}\{M \in da\}] = 2 \int_a^\infty (z - a)z\phi(z) dz da = 2\bar{\Phi}(a) da.$$

Hence $\mathbb{E}[(\eta - M)^+(M - D)] = 2 \int_0^\infty \mathbb{E}[(\eta - a)^+] \bar{\Phi}(a) da$. Write $\bar{\Phi}(a) = P(Z'' > a)$ for an independent standard normal Z'' . The expectation becomes

$$2\mathbb{E}[\mathbf{1}\{\eta > 0, Z'' > 0\}\{\eta\mu - \mu^2/2\}], \quad \mu = \min(\eta, Z'').$$

The polar coordinates used for T_1 evaluate this to $\rho^2/4 - T_1/2$. Therefore $\mathbb{E}[RS] - 1 = \rho^2/2 - T_1$.

(e) $\mathbb{E}[GK] = \frac{1}{2}(4 \ln 2 + \Delta_R) - (2 \ln 2 - 1)(1 + \rho^2)$, by (b) and (c).

(f) $P(\eta > M) = P(\rho Z > |Z'|) = \alpha/\pi$, the angle of the cone divided by 2π . The event $\{\eta < m\}$ is disjoint from it and has the same probability.

(g) The loadings follow from (b)–(e) and the small-error limit of (c):

- Parkinson: $\Delta_R/(4 \ln 2) \sim \rho^2/2$;
- Garman–Klass: $\ln 2 \rho^2 - (2 \ln 2 - 1)\rho^2 = (1 - \ln 2)\rho^2$;
- Rogers–Satchell: $\rho^2/2 + O(\rho^3)$;
- the daily Yang–Zhang form: $\rho^2\{1 + k + (1 - k)/2\} = \frac{3+k}{2}\rho^2$.

□

B.3 Proof of Proposition 3

(a) By the tower property, $\mathbb{E}[g(o_L)r] = \mathbb{E}[g(o_L)\mathbb{E}[r | o_L]] = b_L\mathbb{E}[g(o_L)o_L]$. For the clip function, $g(y)\{y - g(y)\}$ is:

- 0 if $|y| \leq \beta$;
- $\beta(y - \beta) > 0$ if $y > \beta$;
- $-\beta(y + \beta) > 0$ if $y < -\beta$.

Hence $\mathbb{E}[g(o_L)o_L] \geq \mathbb{E}[g(o_L)^2]$, with equality if and only if $P(|o_L| > \beta) = 0$. So $F \geq 1$, and $b = b_L F \geq b_L$ when $b_L \geq 0$.

(b) On $A = \{|o_L| < \beta\}$, $g(o_L) = o_L$ and $\mathbb{E}[o_L r \mathbf{1}_A] = b_L \mathbb{E}[o_L^2 \mathbf{1}_A]$. On A^c , $g(o_L) = \beta \text{sign}(o_L)$. So $\mathbb{E}[g(o_L)r \mathbf{1}_{A^c}] = b_L \beta \mathbb{E}[|o_L| \mathbf{1}_{A^c}]$ and $\mathbb{E}[g(o_L)^2 \mathbf{1}_{A^c}] = \beta^2 P(A^c)$. Since $\mathbb{E}[|o_L| | A^c] \geq \beta$, the pinned coefficient is at least b_L when $b_L \geq 0$.

(c) Let $z = o_L/\sigma$. The function g is Lipschitz with $g'(z) = \mathbf{1}\{|z| < k\}$ almost everywhere, so Stein's identity (Stein, 1981) gives $\mathbb{E}[g(z)z] = \mathbb{E}[g'(z)] = 2\Phi(k) - 1$. Also

$$\mathbb{E}[g(z)^2] = \mathbb{E}[z^2 \mathbf{1}\{|z| \leq k\}] + k^2 P(|z| > k) = 2\Phi(k) - 1 - 2k\phi(k) + 2k^2 \bar{\Phi}(k).$$

These two expectations have integral forms:

$$N(k) := 2\Phi(k) - 1 = \int_0^k 2\phi(t) dt, \quad D(k) := \mathbb{E}[\min(z^2, k^2)] = \int_0^k 4t \bar{\Phi}(t) dt.$$

Lemma B.1. *Let $a, c > 0$ be continuous on $(0, \infty)$, with a/c strictly decreasing. Then $k \mapsto \int_0^k a / \int_0^k c$ is strictly decreasing.*

Proof. Write $A(k) = \int_0^k a$ and $C(k) = \int_0^k c$. The derivative of A/C has the sign of $a(k)C(k) - c(k)A(k) = c(k) \int_0^k c(t) \{a(k)/c(k) - a(t)/c(t)\} dt$. This is negative because the integrand is negative for $t < k$. □

Here $a(t)/c(t) = \lambda(t)/(2t)$, with $\lambda = \phi/\bar{\Phi}$. Since $\lambda' = \lambda(\lambda - t)$,

$$(\lambda/t)' = \lambda(t\lambda - t^2 - 1)/t^2.$$

This is negative if and only if $\lambda(t) < t + 1/t$, which is Gordon's (1941) inequality $\bar{\Phi}(t) > t\phi(t)/(1 + t^2)$. Lemma B.1 therefore makes $F = N/D$ strictly decreasing. The limits follow from:

- $N(k), D(k) \rightarrow 1$ as $k \rightarrow \infty$;
- $N(k) = 2\phi(0)k + O(k^3)$ and $D(k) = k^2 + O(k^3)$ as $k \rightarrow 0$.

The pinned ratio is $\mathbb{E}[|z| \mid |z| > k]/k = \lambda(k)/k$.

(d) The decomposition is a telescoping identity. Under Assumption 3:

- in regime 0, $b_0/b_0^{\text{int}} = F_0$ by (a) and (b);
- in regime 1, $b_1^{\text{int}}/b_1 = 1/F_1 \leq 1$.

□

B.4 Proof of Proposition 4

(a) For any $c > 0$,

$$\text{QL}(Y, f) - \text{QL}(Y, cf) = \frac{Y}{f} \left(1 - \frac{1}{c}\right) - \ln c.$$

Average over security i 's forecasts and put $c = c_i^* = \mathbb{E}_i[Y/f]$. The result is $c_i^* - 1 - \ln c_i^* = q(c_i^*)$, which is non-negative because $\ln c \leq c - 1$. The same algebra holds for sample means. Moreover, c_i^* minimises $c \mapsto \mathbb{E}_i[\text{QL}(Y, cf)]$, whose derivative is $1/c - \mathbb{E}_i[Y/f]/c^2$. So the first term is the loss after the best rescaling.

(b) The pooled loss of κf is $\sum_i w_i \mathbb{E}_i[\text{QL}(Y, \kappa f)]$. Its derivative in κ vanishes when $\sum_i w_i \mathbb{E}_i[Y/(\kappa f)] = 1$. The second-order form follows from $q(c) = \frac{1}{2}(c - 1)^2 + O((c - 1)^3)$ and $\ln c = (c - 1) + O((c - 1)^2)$.

(c) The three expectations are:

- $\mathbb{E}[(\beta o)^2] = \beta^2(\mathbb{E}[o^{*2}] + \sigma_\eta^2)$, by independence;
- $\mathbb{E}[R^2]/(4 \ln 2) = \sigma_c^2 \{1 + \Delta_R(\rho)/(4 \ln 2)\}$, by Proposition 2(c);
- $\mathbb{E}[r^2] = \mathbb{E}[o^{*2}] + \sigma_c^2$.

Divide the sum of the first two by the third. The first-order form uses $\Delta_R/(4 \ln 2) = \rho^2/2 + o(\rho^2)$.

(d), *error-only heterogeneity*. Put $s_i \equiv s$. Then $\psi_i = a + \gamma n_i$, with $a = 1 - (1 - \beta^2)s$ and $\gamma = \beta^2 + \frac{1}{2}$, so $\text{Var}(\ln \psi_i) = \text{Var}(\ln(a/\gamma + n_i))$. For a non-degenerate $n \geq 0$, the map $\xi \mapsto \text{Var}(\ln(\xi + n))$ is strictly decreasing on $\xi > 0$. Its derivative is $2 \text{Cov}(\ln(\xi + n), 1/(\xi + n)) < 0$, because the two variables are strictly oppositely monotone in n . Also,

$$\frac{d(a/\gamma)}{d(\beta^2)} = \frac{s\gamma - a}{\gamma^2} = \frac{\frac{3}{2}s - 1}{\gamma^2},$$

which is negative when $s < 2/3$. So the variance increases with β .

(d), *overnight-share-only heterogeneity*. Put $n_i \equiv n$. Then $\psi_i = (1 + \gamma n)\{1 - t s_i\}$, with $t = (1 - \beta^2)/(1 + \gamma n)$ strictly decreasing in β . The map $t \mapsto \text{Var}(\ln(1 - ts))$ is strictly increasing on $[0, 1/\max s)$. Its derivative is $2 \text{Cov}(\ln(1 - ts), -s/(1 - ts)) > 0$, because both variables are strictly decreasing in s . At $\beta = 1$, $t = 0$ and the variance is zero. □

B.5 Proof of Proposition 5

(a) The window holds L dates. By assumption $R_s/N = \rho_{j(s)}a_s + o_p(1)$ and $A_s/N = a_s + o_p(1)$ for each of them. Since $\sum_s a_s > 0$, the continuous mapping theorem gives

$$\kappa_t = \frac{\sum_s R_s/N}{\sum_s A_s/N} \rightarrow \frac{\sum_s \rho_{j(s)}a_s}{\sum_s a_s} = \omega_t \rho_1 + (1 - \omega_t) \rho_0.$$

(b) If a_s is constant, $\omega_{\tau+j}$ is the share of post-change dates among the L , which is $\min(j + 1, L)/L$.

(c) Write $\ln \kappa_t - \ln \rho_t = \text{bias}_t + \epsilon_t$, with $\mathbb{E}\epsilon_t = 0$ and $\mathbb{E}\epsilon_t^2 = v/L$ to first order. The cross term is zero because the bias is deterministic given the change dates. With probability $1 - O(\lambda L)$, no second change falls in the window after a change at τ . Then, for $0 \leq j < L$,

$$\text{bias}_{\tau+j} = \ln\{(1 - \omega_j)\rho_0 + \omega_j\rho_1\} - \ln \rho_1 = \pm(1 - \omega_j)\delta + O(\delta^2),$$

and the bias is zero afterwards. Summing with the linear weights of (b),

$$\sum_{j=0}^{L-1} (1 - \omega_j)^2 \delta^2 = \delta^2 \frac{(L-1)(2L-1)}{6L} = \frac{\delta^2 L}{3} + O(\delta^2).$$

Changes arrive at rate λ per date, so the long-run average squared bias per date is $\lambda \delta^2 L/3$, up to terms of higher order in δ and λL . Adding the variance gives $\frac{1}{2}(\lambda \delta^2 L/3 + v/L)$. Minimising over L gives $L^* = \sqrt{3v/(\lambda \delta^2)}$, and substituting L^* gives the minimum $\sqrt{\lambda \delta^2 v/3}$. The level loss of a scale error e^ϵ is $q(e^\epsilon) = \epsilon^2/2 + O(\epsilon^3)$, as in Proposition 4(b). $\square$

B.6 Proof of Proposition 6

(a) By iterated expectations, $\mathbb{E}[U_{it} \mid \mathcal{F}_{i,t-1}] = \mathbb{E}[\mathbb{E}[U_{it} \mid IV_{it}, \mathcal{F}_{i,t-1}] \mid \mathcal{F}_{i,t-1}] = 0$. So $\mathbb{E}[X_{k,it} \mid \mathcal{F}_{i,t-1}] = \alpha_{k,i} + \beta_k V_{it}$. For $Z \in \mathcal{F}_{i,t-1}$ within a security, $\text{Cov}(X_k, Z) = \text{Cov}(\mathbb{E}[X_k \mid \mathcal{F}_{t-1}], Z) = \beta_k \text{Cov}(V, Z)$, and the same holds for $k = 0$ with $\beta_0 = 1$. Forward orthogonal deviations remove $\alpha_{k,i}$ while keeping the transformed errors orthogonal to instruments dated $t-1$ (Arellano and Bover, 1995).

(b) With $X = \alpha + \beta IV + U$ and IV uncorrelated with U :

- $\text{Cov}(X_k, X_0) = \beta_k \text{Var}(IV) + \text{Cov}(U_k, U_0)$;
- $\text{Var}(X_0) = \text{Var}(IV) + \text{Var}(U_0)$.

The mean-ratio decomposition is an identity.

(c) In general, $\text{Cov}(X_k, Z) = \beta_k \text{Cov}(IV, Z) + \text{Cov}(U_k, Z)$. Also $\text{Cov}(IV, Z) = \text{Cov}(V, Z)$, because $IV - V$ is uncorrelated with $\mathcal{F}_{t-1}$. If U is a moving average of order q with respect to $\mathcal{F}$, then $\mathbb{E}[U_t \mid \mathcal{F}_{t-q-1}] = 0$, so instruments in $\mathcal{F}_{t-q-1}$ are uncorrelated with U_t .

(d) These are the standard results for generalised method of moments estimators (Hansen, 1982), applied to the moment conditions of (a) after the transformation.

(e) Minimising $w' \Sigma_X w$ subject to $w' \beta = 1$ gives $w \propto \Sigma_X^{-1} \beta$. By the Sherman–Morrison formula,

$$\Sigma_X^{-1} \beta = \frac{\Omega_U^{-1} \beta}{1 + \text{Var}(IV) \beta' \Omega_U^{-1} \beta} \propto \Omega_U^{-1} \beta.$$

Equivalently, under the constraint, $w' \Sigma_X w = \text{Var}(IV) + w' \Omega_U w$. $\square$

B.7 Proof of Proposition 7

(a) By Proposition 1(a), $\mathbb{E}[oo^*] = b\mathbb{E}[o^2]$. Hence

$$\mathbb{E}[(o^* - bo)^2] = \mathbb{E}[o^{*2}] - 2b\mathbb{E}[oo^*] + b^2\mathbb{E}[o^2] = \mathbb{E}[o^{*2}] - b^2\mathbb{E}[o^2],$$

which is the identity. The deficit is zero if and only if $o^* = bo$ almost surely.

- If $\mathbb{E}[o^* \eta] = 0$, then $\mathbb{E}[oo^*] = \mathbb{E}[o^{*2}]$, so $b\mathbb{E}[o^2] = \mathbb{E}[o^{*2}]$ and $b^2\mathbb{E}[o^2] = b\mathbb{E}[o^{*2}]$. Also $\mathbb{E}[\pi(o)] = b^2\mathbb{E}[o^2] + b(1-b)\mathbb{E}[o^2] = b\mathbb{E}[o^2] = \mathbb{E}[o^{*2}]$.
- If, in addition, (o^*, η) is centred Gaussian, then so is (o^*, o) . Hence $\mathbb{E}[o^* \mid o] = bo$, and $\text{Var}(o^* \mid o) = \mathbb{E}[o^{*2}] - b^2\mathbb{E}[o^2] = b(1-b)\mathbb{E}[o^2]$ is constant. So $\mathbb{E}[o^{*2} \mid o] = b^2 o^2 + b(1-b)\mathbb{E}[o^2] = \pi(o)$.
- The last statement is Proposition 1(e). The deficit $\mathbb{E}[o^{*2}] - b^2\mathbb{E}[o^2]$ takes every value in $[0, \mathbb{E}[r^2] - b^2\mathbb{E}[o^2]]$ as $\mathbb{E}[o^{*2}]$ ranges over the identified set.

(b) The open lies in $[l, h]$ and the previous close is the point 0. There are three cases:

- if $bo \in [l, h]$, both extensions vanish and $R^* = h - l$;
- if $bo > h$, then $R^* = h - l + (bo - h) = bo - l$;

- if $bo < l$, then $R^* = h - bo$.

In each case $R^* = \max(h, bo) - \min(l, bo)$, the range of the set $\{l, h, bo\}$. For $b \in [0, 1]$, bo is a convex combination of 0 and $o \in [l, h]$. So $\{l, h, bo\}$ lies in the interval spanned by $\{l, h, 0\}$, whose length is TR ; and it contains $\{l, h\}$, whose span is R . At $b = 1$, $bo = o \in [l, h]$. In the example, $h = \ln(O/C_-)$, $l = r = 0$ and $b = 0$, so $R^* = TR = h$ and $A_0 = (1 - \lambda_0)h^2/(4 \ln 2)$.

(c), *the true range*. Take $\sigma_c = 1$. Measured from C_- , the printed high and low are $h = o^* + \max(\eta, M)$ and $l = o^* + \min(\eta, m)$. Without the error they are $h_e = o^* + M$ and $l_e = o^* + m$. Write $x^+ = \max(x, 0)$ and $x^- = \max(-x, 0)$. Then $TR = h^+ + l^-$, so $TR - TR_e = \delta^+ + \delta^-$, where $\delta^+ = h^+ - h_e^+ \geq 0$ and $\delta^- = l^- - l_e^- \geq 0$. Hence

$$TR^2 - TR_e^2 = 2TR_e(\delta^+ + \delta^-) + (\delta^+ + \delta^-)^2.$$

The second term is at most $\eta^2 \mathbf{1}\{M < |\eta|\}$ or $\eta^2 \mathbf{1}\{-m < |\eta|\}$. Since $P(M < x) \leq \sqrt{2/\pi} x$, its expectation is $O(\rho^3)$.

Now $\delta^+ > 0$ requires $\eta > M$. There are two cases.

- On $\{o^* > 0\}$, $\delta^+ = (\eta - M)^+$.
- On $\{o^* \leq 0\}$, $\delta^+ \leq \eta^+ \mathbf{1}\{|o^*| < \eta\}$. The probability that $|o^*| < \eta$ and $M < \eta$, given η , is $O(\eta^2)$, so this case contributes $O(\rho^3)$.

Conditioning on $\eta = e > 0$, for small e ,

$$\begin{aligned} \mathbb{E}[TR_e(e - M)^+ \mathbf{1}\{o^* > 0\}] &= \int_0^e (e - x) \mathbb{E}[TR_e \mathbf{1}\{o^* > 0\} \mid M = x] f_M(x) dx \\ &= \frac{e^2}{2} f_M(0) \mathbb{E}[TR_e \mathbf{1}\{o^* > 0\} \mid M = 0^+] \{1 + o(1)\}, \end{aligned}$$

with $f_M(0) = \sqrt{2/\pi}$. Conditional on $M = 0^+$:

- the path $-W$ is a Brownian meander (Durrett et al., 1977), so $-m$ has the law of its maximum U , independent of o^* ;
- on $\{o^* > 0\}$, $TR_e = o^* - \min(o^* + m, 0) = \max(o^*, -m)$.

Integrating over η , with $\mathbb{E}[(\eta^+)^2] = \rho^2/2$, and adding the mirror image δ^- under $(W, \eta, o^*) \mapsto (-W, -\eta, -o^*)$, gives

$$\mathbb{E}[TR^2 - TR_e^2] = \frac{2\rho^2}{\sqrt{2\pi}} \mathbb{E}[\max(o^*, U) \mathbf{1}\{o^* > 0\}] + o(\rho^2).$$

Dominated convergence applies as in the small-error limit of B.2(c). Next,

$$\mathbb{E}[\max(o^*, U) \mathbf{1}\{o^* > 0\}] = \frac{1}{2} \mathbb{E}[U] + \mathbb{E}[(o^* - U)^+] = \frac{1}{2} \sqrt{2\pi} \ln 2 + s \mathbb{E}[\psi(U/s)].$$

This uses $\mathbb{E}[(sZ - u)^+] = s\psi(u/s)$ for $u \geq 0$, and $\mathbb{E}[U] = \sqrt{2\pi} \ln 2$. The mean follows from the law of U (Durrett and Iglehart, 1977):

$$\mathbb{E}[U] = \int_0^\infty P(U > u) du = 2\sqrt{\pi/2} \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} = \sqrt{2\pi} \ln 2.$$

Dividing by $4 \ln 2$ gives (7).

(c), *the properties of γ_{TR}* .

- *Monotone*. Since $\psi'(a) = -\bar{\Phi}(a)$, $\frac{d}{ds}\{s\psi(u/s)\} = \psi(u/s) + (u/s)\bar{\Phi}(u/s) = \phi(u/s) > 0$.
- *Upper bound*. $0 \leq \psi(a) \leq \psi(0) = 1/\sqrt{2\pi}$ gives $\frac{1}{4} \leq \gamma_{TR}(s) \leq \frac{1}{4} + s/(4\pi \ln 2)$.
- *Lower bound*. Since $\bar{\Phi} \leq \frac{1}{2}$ on $[0, \infty)$, $\psi(a) \geq 1/\sqrt{2\pi} - a/2$. So $s\mathbb{E}[\psi(U/s)] \geq s/\sqrt{2\pi} - \frac{1}{2}\mathbb{E}[U]$, and hence $\gamma_{TR}(s) \geq s/(4\pi \ln 2)$.

- *Limits.* As $s \rightarrow 0$, $s\psi(U/s) \leq s/\sqrt{2\pi} \rightarrow 0$. As $s \rightarrow \infty$, $0 \leq s\psi(U/s) - s/\sqrt{2\pi} + U/2 = s \int_0^{U/s} \{\Phi(z) - \frac{1}{2}\} dz \leq \phi(0)U^2/(2s)$, using $\Phi(z) - \frac{1}{2} \leq z\phi(0)$. The bound tends to zero because $\mathbb{E}[U^2] < \infty$.

(c), *the kernels.* Under Assumption 2 the close carries no error, so $\lambda_0 r^2$ is the same with and without it, and A_0 has loading $(1 - \lambda_0)\gamma_{TR}(s)$.

For the full kernel:

- $b = s^2/(s^2 + \rho^2) = 1 - \rho^2/s^2 + O(\rho^4)$ and $w = \lambda_0 \rho^2/s^2 + O(\rho^4)$;
- $\mathbb{E}[(bo)^2] = b^2(s^2 + \rho^2) = s^2 - \rho^2 + O(\rho^4)$.

The anchor bo lies within $(1 - b)|o| = O(\rho^2)|o|$ of the printed open. So $R^* \neq R$ only if the open is an extreme or within that distance of one, an event of probability $O(\rho)$, and there $R^* - R = O(\rho^2)|o|$. Hence $\mathbb{E}[R^{*2} - R^2] = O(\rho^3)$, and $\mathbb{E}[R^{*2}]/(4 \ln 2) = 1 + \frac{1}{2}\rho^2 + o(\rho^2)$ by Proposition 2(c). With $\mathbb{E}[r^2] = s^2 + 1$,

$$\mathbb{E}[A] = (1 - w)(s^2 - \rho^2 + 1 + \frac{1}{2}\rho^2) + w(s^2 + 1) + o(\rho^2) = s^2 + 1 - \frac{1}{2}\rho^2 + o(\rho^2).$$

At $\sigma_\eta = 0$, $A = o^{*2} + (M - m)^2/(4 \ln 2)$, with mean $s^2 + 1$. So the loading is $-\frac{1}{2}$. With $\pi(o)$ in place of $(bo)^2$, $\mathbb{E}[\pi(o)] = s^2$ exactly by (a). The same steps then give $s^2 + 1 + \frac{1}{2}\rho^2 + o(\rho^2)$, a loading of $+\frac{1}{2}$. $\square$

B.8 Proof of Proposition 8

(a) By definition $m_{-i} = (N - 1)^{-1} \sum_{j \neq i} (\beta_j f + \zeta_j + \eta_j)$, a measurable function of f and $(\zeta_j, \eta_j)_{j \neq i}$. By Assumption 4 these are independent of (ζ_i, η_i) . Security i 's print o_i does not enter the sum.

(b) $m_{-i} - \bar{\beta}_{-i}f = (N - 1)^{-1} \sum_{j \neq i} (\zeta_j + \eta_j)$. The summands are independent with mean zero, so the second moment of the sum is the sum of the second moments. Bounding each by the largest gives the inequality.

(c) Write $\epsilon_{-i} = m_{-i} - \bar{\beta}_{-i}f$. Since ϵ_{-i} has mean zero and is independent of f , $\mathbb{E}[m_{-i}^2] = \bar{\beta}_{-i}^2 \sigma_f^2 + \mathbb{E}[\epsilon_{-i}^2]$, and the second term is $V_{N,i}$ by (b). Next,

$$\mathbb{E}[r_i m_{-i}] = \mathbb{E}[(\beta_i f + \zeta_i + x_i)(\bar{\beta}_{-i}f + \epsilon_{-i})] = \beta_i \bar{\beta}_{-i} \sigma_f^2.$$

Three terms vanish:

- $\mathbb{E}[\zeta_i \epsilon_{-i}] = 0$ and $\mathbb{E}[\zeta_i f] = 0$, by independence and zero means;
- $\mathbb{E}[\beta_i f \epsilon_{-i}] = 0$, by the same argument;
- $\mathbb{E}[x_i m_{-i}] = 0$, because m_{-i} is a function of what is known at the open, by the last clause of the assumption.

Summing over i and dividing gives b_M . With equal betas $\bar{\beta}_{-i} = \beta$, and the ratio tends to one because $V_N \rightarrow 0$ by (b).

(d) By the same steps, $\mathbb{E}[o_i r_i] = \beta_i^2 \sigma_f^2 + \mathbb{E}[\zeta_i^2] + \mathbb{E}[\zeta_i \eta_i]$, using $\mathbb{E}[o_i x_i] = 0$, and $\mathbb{E}[o_i^2] = \beta_i^2 \sigma_f^2 + \mathbb{E}[(\zeta_i + \eta_i)^2]$. Write $S_M = \sum_i \beta_i^2 \sigma_f^2$ and $S_I = \sum_i \mathbb{E}[(\zeta_i + \eta_i)^2]$. Then

$$b = \frac{\sum_i \mathbb{E}[o_i r_i]}{\sum_i \mathbb{E}[o_i^2]} = \frac{S_M + b_I S_I}{S_M + S_I} = \mu + (1 - \mu) b_I.$$

In a sample, the parts $o_{I,i} = o_i - o_M$ sum to zero within each date, so $\sum o_M o_I = \sum_t o_{M,t} \sum_i o_{I,i,t} = 0$. Hence:

- $\sum o^2 = \sum o_M^2 + \sum o_I^2$;
- $\sum o r = \sum o_M r + \sum o_I r = \hat{b}_M \sum o_M^2 + \hat{b}_I \sum o_I^2$.

Dividing the second line by the first gives the identity.

(e) If $o_i \neq 0$, then $o^* = 0$, so $R^* = \max(h, 0) - \min(l, 0) = TR$ and the overnight term vanishes. On a bar with a range, $D = 4 \ln 2$, and $A = (1 - w)TR^2/(4 \ln 2) + wr^2$, the open-free kernel.

If the bar never left the previous close, then $o = c = u = d = 0$ and so $h = l = r = 0$. The open printed at the previous close, so $o^* = m_{-i}$. Then $R^* = |m_{-i}|$ and $D = 1$, so $A = (1 - w)(m_{-i}^2 + m_{-i}^2)$. Its expectation follows from (c). The open-free kernel is $(1 - \lambda_0) \cdot 0 + \lambda_0 \cdot 0 = 0$.

(f) We take each property in turn.

- With one observation, $R_1 = \max(0, W(1)) - \min(0, W(1)) = |W(1)|$, so $\lambda_1 = \mathbb{E}[W(1)^2] = 1$.
- The grid $\{j/(kn)\}$ contains $\{j/n\}$. A maximum over more points is no smaller, and a minimum no larger, so $R_{kn} \geq R_n$ pathwise.
- Every discrete range is at most the continuous range $\sup_{[0,1]} W - \inf_{[0,1]} W$, which includes $W(0) = 0$. That range has second moment $4 \ln 2$ (Parkinson, 1980).
- Paths are continuous, so the discrete maximum and minimum converge to the supremum and infimum as the mesh $1/n \rightarrow 0$. The squares are dominated by the square of the continuous range, which is integrable, so $\lambda_n \rightarrow 4 \ln 2$ by dominated convergence. $\square$

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